

Problem-Solution Fit

Date	26 June 2026
Team ID	
Project Name	Rising Waters – A Flood Prediction
Maximum Marks	5 Marks

Problem-Solution Fit:

1. CUSTOMER SEGMENT(S) [CS] - Residents in flood-prone areas - Disaster Management Authorities - Local Government Bodies - Farmers & Agricultural Communities - Emergency Response Teams	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Limited internet connectivity - Lack of technical knowledge - Incomplete weather information - Limited access to smart devices - Financial constraints	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Pros: - Easily accessible - Provides general weather updates - Covers large geographical regions Cons: - Limited local-level flood prediction - Delayed alerts - Lower prediction accuracy - Difficult for common users to interpret
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Frequent flood-related property damage - Delayed flood warnings - Inaccurate prediction using traditional methods - Loss of crops and livestock - High risk during heavy rainfall seasons	9. PROBLEM ROOT / CAUSE [RC] - Climate change causing extreme rainfall - Poor drainage systems - River overflow - Urbanization and deforestation - Lack of accurate early-warning systems	7. BEHAVIOR + ITS INTENSITY [BE] - Regularly checks weather forecasts - Searches for flood warnings on news and websites - Shares alerts with family and neighbors - Takes precautionary measures only after official alerts - Depends heavily on government notifications
3. TRIGGERS TO ACT [TR] - Heavy rainfall forecast - Rising river water levels - Government weather alerts - Seasonal monsoon conditions - Need for early evacuation planning	10. YOUR SOLUTION [SL] Develop an intelligent Flood Prediction System using Machine Learning algorithms (Decision Tree, Random Forest, KNN, and XGBoost) integrated with a Flask web application. The system analyzes weather and environmental parameters to accurately predict flood risk, providing early warnings that help authorities and citizens take preventive measures, reduce losses, and improve disaster preparedness.	8. CHANNELS OF BEHAVIOR [CH] ONLINE - Flood Prediction Web Application - Government Weather Portals - Mobile Applications - News Websites - Social Media Alerts OFFLINE - Television News - Radio Broadcasts - Local Government Announcements - Public Warning Sirens - Community Awareness Programs
4. EMOTIONS BEFORE / AFTER [EM] Before: Fear, Anxiety, Confusion, Uncertainty, Lack of preparedness After: Confidence, Safety awareness, Better preparedness, Peace of mind, Quick decision making		